

FIITJEE NSEC MOCK TEST-3

Time Allotted: 2 Hours

CODE:

Maximum Marks: 216

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Write the question paper code (mentioned above) on YOUR OMR Answer Sheet (in the space provided), otherwise your Answer Sheet will NOT be evaluated. Note that the same Question Paper Code appears on each page of the question paper.

Instructions to Candidates:

1. Use of mobile phone, smart watch, and ipad during examination is STRICTLY PROHIBITED.
2. In addition to this question paper, you are given OMR Answer Sheet along with Candidate's copy.
3. On the OMR sheet, make all the entries carefully in the space provided **ONLY** in **BLOCK CAPITALS** as well as by properly darkening the appropriate bubbles.

Incomplete/ incorrect/ carelessly filled information may disqualify your candidature.

4. On the OMR Answer Sheet, use only **BLUE** or **BLACK BALLPOINT PEN** for making entries and filling the bubbles.
5. **Your Enrollment number entered** on the OMR Answer Sheet shall remain your login credentials means login id and password respectively for accessing your performance / result in NSE.
6. Question paper has two parts. In part A1 (Q. No.1 to 48) each question has four alternatives, out of which only one is correct. Choose the correct alternative (s) and fill the appropriate bubble (s), as shown.

Q. No. 22	☐ a	☒	☐ c	☐ d
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In part A2 (Q. No. 49 to 60) each question has four alternatives out of which any number of alternative (s) (1, 2, 3, or 4) may be correct. You have to choose all correct alternative(s) and fill the appropriate bubble(s), as shown

Q. No. 54	☐ a	☒	☐ c	☒
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7. For **Part A1**, each correct answer carries 3 marks whereas 1 mark will be deducted for each wrong answer. In **Part A2**, you get 6 marks if all the correct alternatives are marked. No negative marks in this part.
8. Rough work may be done in the space provided. There are 15 printed pages in this paper
9. Use of **non - programmable scientific** calculator is allowed.
10. No candidate should leave the examination hall before the completion of the examination.
11. After submitting answer paper, take away the question paper & Candidate's copy OMR sheet for your reference.

Please DO NOT make any mark other than filling the appropriate bubbles properly in the space provided on the OMR answer sheet.

OMR answer sheets are evaluated using machine, hence CHANGE OF ENTRY IS NOT ALLOWED. Scratching or overwriting may result in a wrong score.

DO NOT WRITE ON THE BACK SIDE OF THE OMR ANSWER SHEET.

Name of the Candidate : _____

Batch : _____ Date of Examination : _____

Enrolment Number : _____

**INDIAN ASSOCIATION OF PHYSICS TEACHERS
 NATIONAL STANDARD EXAMINATION IN CHEMISTRY
 (NSEC-2023)**

Time: 120 minutes

Max. Marks: 216

Attempt All Sixty Questions

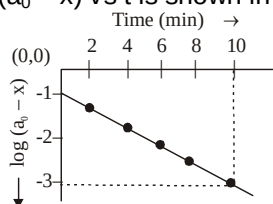
A – 1

ONLY ONE OUT OF FOUR OPTIONS IS CORRECT BUBBLE THE CORRECTION OPTION

1. A mixture of H_2 and SO_2 gas in a mass ratio of 1: $\sqrt{2}$ was taken and subjected to effusion from the same vessel. The mass of the gas effused is $z \times 0.4$; if total 1.29 moles have effused. What is the value of z ?

(A) 6 (B) 8
(C) 4 (D) 2

2. For the first order decomposition of $SO_2Cl_2(g)$,
 $SO_2Cl_2(g) \longrightarrow SO_2(g) + Cl_2(g)$
 a graph of $\log(a_0 - x)$ vs t is shown in figure. What is the rate constant (sec^{-1})?

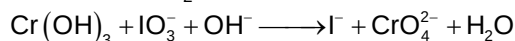


(A) 0.2 (B) 4.6×10^{-1}
(C) 7.7×10^{-3} (D) 1.15×10^{-2}

3. Find the number of non polar molecule having all polar bonds.
 CCl_4 , PCl_5 , SF_6 , IF_7 , IF_5 , PCl_4F , SF_4 , NO_2 , BF_3

(A) 3 (B) 11
(C) 9 (D) 5

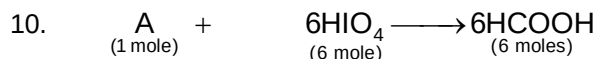
4. When the following redox reaction is balanced using simplest 'natural numbers', the stoichiometric coefficient of H_2O will be:



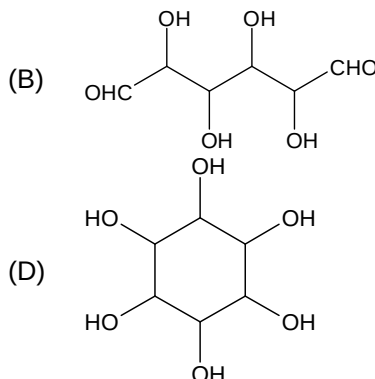
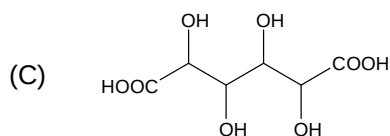
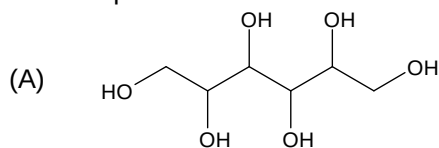
(A) 2 (B) 3
(C) 5 (D) it is impossible to balance this reaction

Space for rough work

5. Which of the following about SF_4 , SOF_4 , CH_2SF_4 and OCF_2 molecules is correct?
 (A) Equatorial FSF bond angle in SOF_4 will be less than in SF_4 molecule
 (B) The two hydrogens, carbon, sulphur and two fluorines in molecule CH_2SF_4 will be lying in the same plane
 (C) The bond angle FCO will be $< 120^\circ$ in molecule OCF_2
 (D) The axial FSF bond angle in $\text{SF}_4 = 90^\circ$
6. Select the correct option if it is known that $K_{\text{sp}}(\text{AgCl}) > K_{\text{sp}}(\text{AgBr}) > K_{\text{sp}}(\text{AgI})$
 (A) $E^\circ_{\text{I}^-|\text{AgI}|\text{Ag}} > E^\circ_{\text{Br}^-|\text{AgBr}|\text{Ag}} > E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}}$ (B) $E^\circ_{\text{I}^-|\text{AgI}|\text{Ag}} < E^\circ_{\text{Br}^-|\text{AgBr}|\text{Ag}} < E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}}$
 (C) $E^\circ_{\text{I}^-|\text{AgI}|\text{Ag}} < E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}} < E^\circ_{\text{Br}^-|\text{AgBr}|\text{Ag}}$ (D) $E^\circ_{\text{I}^-|\text{AgI}|\text{Ag}} = E^\circ_{\text{Br}^-|\text{AgBr}|\text{Ag}} = E^\circ_{\text{Cl}^-|\text{AgCl}|\text{Ag}}$
7. A mixture of hydrocarbon containing acetylene (A) and ethane (B) when burned under controlled conditions produced 16.20 g of water and 139.7 kcal of heat. Standard heats of combustion for acetylene and ethane are respectively -310.6 and $-372.8 \text{ kcal mol}^{-1}$. What is the composition of the mixture?
 (A) (A) = 0.25 mol and (B) = 0.15 mol (B) (A) = 0.15 mol and (B) = 0.25 mol
 (C) (A) = 0.25 mol and (B) = 0.10 mol (D) (A) = 0.10 mol and (B) = 0.25 mol
8. 1 M NaCl and 1M HCl are present in an aqueous solution. The solution is
 (A) not a buffer solution and with $\text{pH} < 7$
 (B) not a buffer solution with $\text{pH} > 7$
 (C) a buffer solution with $\text{pH} < 7$
 (D) a buffer solution with $\text{pH} > 7$
9. An iron cylinder contains helium at a pressure of 250 kPa at 300 K. The cylinder can withstand a pressure of $1 \times 10^6 \text{ Pa}$. The room in which cylinder is placed catches fire. (M.P. of cylinder = 1800 K). Which of the following statement is true?
 (A) At 1800 K the pressure inside the cylinder is $5 \times 10^5 \text{ Pa}$
 (B) The cylinder will not burst before melting
 (C) The cylinder will burst before melting
 (D) Helium gas will solidify at 1800 K



The compound A is



Space for rough work

11. For the decomposition reaction
 $\text{NH}_2\text{COONH}_4(\text{s}) \rightleftharpoons 2\text{NH}_3(\text{g}) + \text{CO}_2(\text{g})$
 The $K_p = 2.9 \times 10^{-5} \text{ atm}^3$. The total pressure of gases at equilibrium when 1 mole of $\text{NH}_2\text{COONH}_4(\text{s})$ was taken to start with would be
 (A) 0.0194 atm (B) 0.0388 atm
 (C) 0.0582 atm (D) 0.0766 atm

12. The compound which has one isopropyl group is:
 (A) 2,2,3,3-Tetramethyl pentane (B) 2,2-Dimethyl pentane
 (C) 2,2,3-Trimethyl pentane (D) 2-Methyl pentane

13. 200 ml of 0.1 M NaOH is mixed with 300 ml of 0.2 M $\text{Ba}(\text{OH})_2$ then the resulting solution has
 (A) molarity = 1.6 M (B) normality = 0.28 N
 (C) normality = 1.6 N (D) molarity = 3.2 N

14. Roasted gold ore + $\text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} [\text{X}] + \text{OH}^-$,
 $[\text{X}] + \text{Zn} \longrightarrow [\text{Y}] + \text{Au}$
 $[\text{X}]$ and $[\text{Y}]$ are :
 (A) $\text{X} = [\text{Au}(\text{CN})_2]^-$; $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$ (B) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$; $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
 (C) $\text{X} = [\text{Au}(\text{CN})_2]^-$; $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$ (D) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$; $\text{Y} = [\text{Zn}(\text{CN})_6]^{2-}$

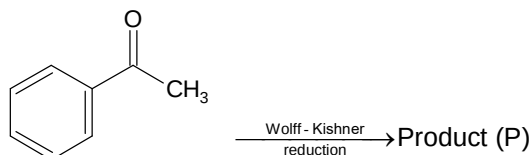
15. The rate law for a reaction between the substances A and B is given by:
 $\text{Rate} = k [\text{A}]^n [\text{B}]^m$
 On doubling the concentration of A and halving the concentration of B, the ratio of the new rate to the earlier rate of the reaction will be as
 (A) $2^{(n-m)}$ (B) $1/2^{(m+n)}$
 (C) $(m+n)$ (D) $(n-m)$

16. Heats of combustion of $\text{CH}_4(\text{g})$, $\text{C}_2\text{H}_4(\text{g})$ and $\text{C}_2\text{H}_6(\text{g})$ are -890 , -1411 and -1560 kJ / mole ; most favourable fuel gas combination will be as
 (A) 1: 3: 6 (B) 3: 1: 6
 (C) 6: 3: 1 (D) 8: 1: 1

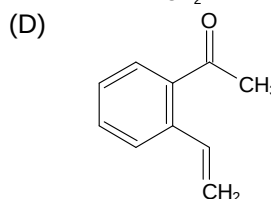
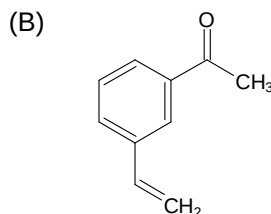
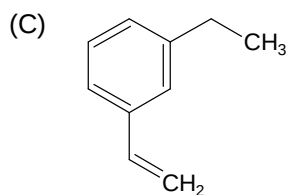
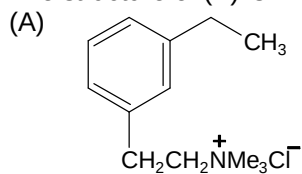
17. The vapour pressure of solvent decreased by 10 mm of mercury when a non-volatile solute was added to the solvent. The mole fraction of solute in the solution is 0.2. What should be mole fraction of solvent if the decrease in the vapour pressure is to be 20 mm of mercury?
 (A) 0.8 (B) 0.6
 (C) 0.4 (D) 0.2

Space for rough work

18.



The structure of (P) is



19.

$$P_A = (235y - 125xy) \text{ mm of Hg}$$

where P_A is partial pressure of A

x is mole fraction of B in liquid phase in the mixture of two liquids A and B and y is mole fraction of A in vapour phase, then P_B° in mm of Hg is

(A) 235

(B) 0

(C) 125

(D) 110

20.

What is the pH value at which $\text{Mg}(\text{OH})_2$ begins to precipitate from a solution containing 0.2 M Mg^{2+} ions? K_{sp} of $\text{Mg}(\text{OH})_2$ is $2 \times 10^{-13} \text{ M}^3$?

(A) 8

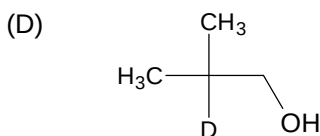
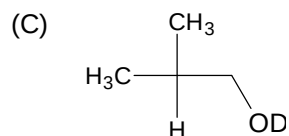
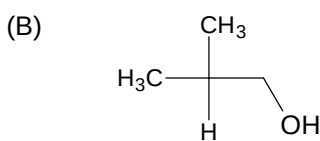
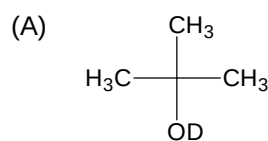
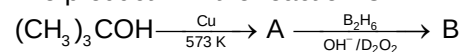
(B) 9

(C) 6

(D) 7

21.

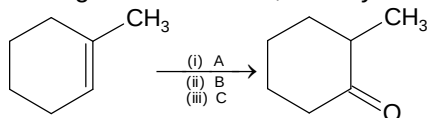
The product B in the reaction is



Space for rough work

22. Consider the following statements for monosaccharides
 (1) They are optically active polyhydroxy carbonyl compound
 (2) Fructose is a ketose sugar and hence it does not give red precipitate with Fehling solution
 (3) α - D - glucose and β - D - glucose are anomer
 Which one of the above statements are corrects
 (A) 1 & 2 (B) 1, 2 & 3
 (C) 1 and 3 (D) 2 & 3

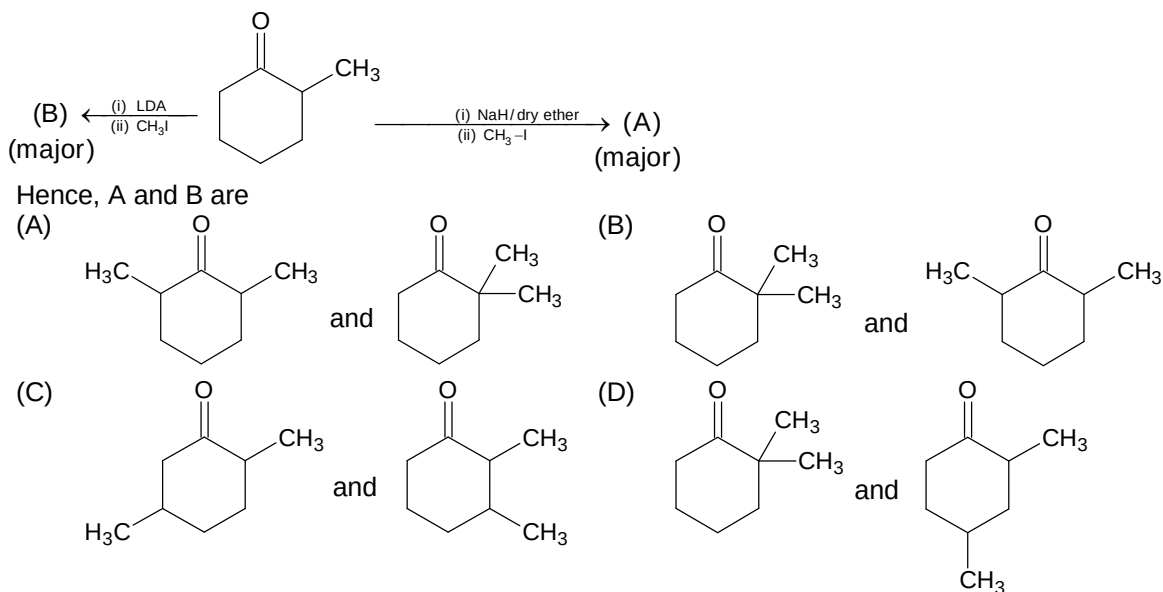
23. In the given conversion, identify the reagents A, B and C.



- (A) A : H_2O_2 in aqueous NaOH, B : PCC/ CH_2Cl_2 , C : $\text{B}_2\text{H}_6/\text{THF}$
 (B) A : $\text{B}_2\text{H}_6/\text{THF}$, B : $\text{Br}_2/\text{CH}_2\text{Cl}_2$, C : PCC/ CH_2Cl_2
 (C) A : $\text{B}_2\text{H}_6/\text{THF}$, B : $\text{H}_2\text{O}_2/\text{OH}^-$, C : PCC/ CH_2Cl_2
 (D) A : $\text{Br}_2/\text{CH}_2\text{Cl}_2$, B : $\text{B}_2\text{H}_6/\text{THF}$, C : $\text{H}_2\text{O}_2/\text{OH}^-$

24. A solution is prepared by mixing 500 ml of NaOH solution of pH = 12 and 500 ml of CH_3COOH solution of pH = 3. The resultant solution must be [$K_a(\text{CH}_3\text{COOH}) = 1.8 \times 10^{-5}$]
 (A) an acidic buffer
 (B) a basic buffer
 (C) a salt of weak acid and strong base undergoing hydrolysis
 (D) a salt of weak acid and strong base along with more NaOH
25. How far from the nucleus is the electron in a hydrogen atom if it has energy of -0.850 eV ?
 (A) $7.3 \text{ \AA}$ (B) $8.46 \text{ \AA}$
 (C) $4.86 \text{ \AA}$ (D) $1.45 \text{ \AA}$

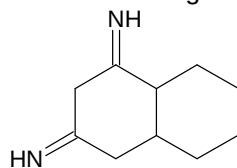
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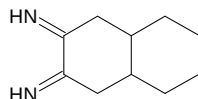
Space for rough work

27. Which of the following imine is least stable?

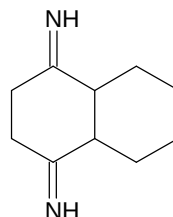
(A)



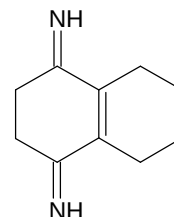
(B)



(C)



(D)



28. If there were three possible values $\left(-\frac{1}{2}, 0, +\frac{1}{2}\right)$ for the spin quantum number s , the number of elements would be there in the fourth period of periodic table are (assume the other quantum numbers are as usual).

(A) 18

(B) 27

(C) 36

(D) 45

29. Which is most basic?

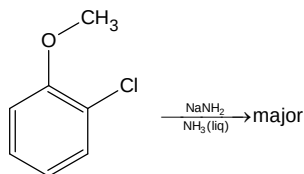
(A) CH_3NCS

(B) SiH_3NCO

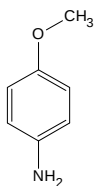
(C) SiH_3NCS

(D) $(\text{SiH}_3)_3\text{N}$

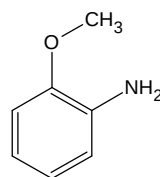
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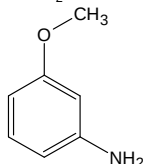
(A)



(B)



(C)



(D) All

31. Three metals A, B and C have work functions as 15.5 eV, 24 eV and 30.5 eV respectively. If the light of wavelength $5000 \text{ \AA}$ is incident on each metal, which of the following is correct?
- (A) Electrons can be ejected from A and C only (B) Electrons can be ejected from A and B only
(C) Electrons can be ejected from A, B and C (D) None of the above

Space for rough work

32. Which one of the following oxoanion contain S – S bond?
 (1) $\text{S}_2\text{O}_8^{-2}$, (2) $\text{S}_2\text{O}_3^{-2}$, (3) $\text{S}_2\text{O}_4^{-2}$, (4) $\text{S}_2\text{O}_6^{-2}$
 Select the correct answers using codes below
 (A) 1, 2 & 3 (B) 1, 3 & 4
 (C) 2, 3 & 4 (D) 2 & 4
33. Anhydrous borax ($\text{Na}_2\text{B}_4\text{O}_7$) is converted in to B_2O_3 by following steps

$$\text{Na}_2\text{B}_4\text{O}_7 \xrightarrow{\text{acid}} \text{H}_3\text{BO}_3 \xrightarrow{\Delta} \text{B}_2\text{O}_3$$

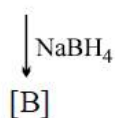
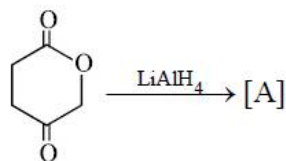
 If 10.1 gm of impure anhydrous borax sample is being treated with acid followed by heating, the amount of B_2O_3 produced 3.5 gm. Find the percentage purity of borax sample
 (A) 100 % pure (B) 50 % pure
 (C) 25 % pure (D) 12.5 % pure
34. The thermal stability of hydrides of group 15 follows the order
 (A) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3 < \text{BiH}_3$ (B) $\text{NH}_3 > \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3$
 (C) $\text{PH}_3 > \text{AsH}_3 > \text{SbH}_3 > \text{BiH}_3 > \text{NH}_3$ (D) $\text{AsH}_3 < \text{PH}_3 > \text{SbH}_3 > \text{NH}_3 > \text{BiH}_3$
35. Among 1-pentene, cis-2-pentene and trans-2-pentene the correct order of stability is
 (A) 1-pentene > cis-2-pentene > trans-2-pentene
 (B) 1-pentene > cis-2-pentene < trans-2-pentene
 (C) 1-pentene < cis-2-pentene > trans-2-pentene
 (D) 1-pentene < cis-2-pentene < trans-2-pentene.
36. In which of the following alcohols C – O bond breaks (heterolytically) with greatest ease:
 (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$ (B) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{CH} - \text{OH} \end{array}$
 (C) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{OH} \\ | \\ \text{CH}_3 \end{array}$ (D) $\begin{array}{c} \text{CH}_3 \\ | \\ \text{CH}_3 - \text{C} - \text{CH}_2 - \text{OH} \\ | \\ \text{CH}_3 \end{array}$
37. The element X, Y, Z and J have the indicated electron configuration starting with the innermost shell.
 The most metallic element is
 (a) X = 2, 8, 3 (b) Y = 2, 8, 8
 (c) Z = 2, 8, 8, 1 (d) J = 2, 8, 8, 7
38. Which has the maximum electron affinity?
 (A) F (B) Cl
 (C) Br (D) I

Space for rough work

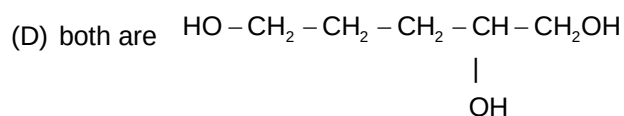
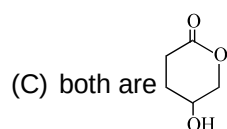
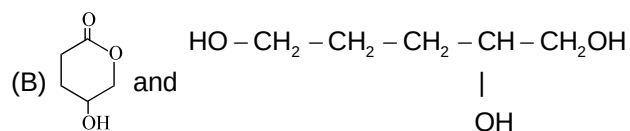
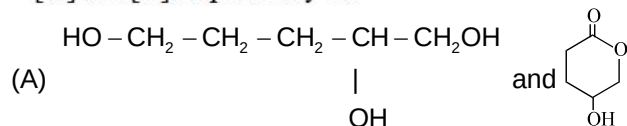
39. Which of the following is (are) true order
(A) $B^+ < B < B^-$ (radii)
(B) $I < Br < Cl < F$ (electron affinity)
(C) $O^- < O^+ < O^{2-}$ (radii)
(D) $Na < Mg < Al < Si$ (Ionisation potential)
40. 100 mL of 1 N NH_4OH ($K_b = 5 \times 10^{-5}$) is neutralized to equivalence point by 1 N HCl. The pH of solution at equivalence point is:
(A) 2.0 (B) 2.5
(C) 3.0 (D) 5.0
41. For the reaction, $PCl_5(g) \rightleftharpoons PCl_3 + Cl_2(g)$, which of the following will not disturb the equilibrium at constant temperature
(A) introducing an inert gas at constant volume (B) introducing an inert gas at constant pressure
(C) increasing the volume of the container (D) introducing PCl_5 at constant volume
42. A vessel with a capacity of 1L contains hydrogen at 300 mm pressure. Another vessel with a capacity of 2L contains oxygen at 600 mm pressure. The two vessels are connected by a valve. If the temperature is kept constant, the final pressure of the gas is
(A) 500 mm (B) 900 mm
(C) 450 mm (D) 300 mm
43. 2L of a gas is at pressure of 2 atm and a temperature of 300 K. What will be the volume of the gas, if the pressure is doubled and the temperature is halved?
(A) 2L (B) 4L
(C) 0.5 L (D) 1L
44. Which of the following complex is an inner d- orbital complex?
(A) $[CoF_6]^{-3}$ (B) $[FeF_6]^{-3}$
(C) $[Cr(NH_3)_6]^{3+}$ (D) $[Co(H_2O)_6]^{2+}$
45. The process of converting hydrated alumina to anhydrous alumina is called
(A) Roasting (B) Smelting
(C) Dressing (D) Calcination
46. The reduction of benzoyl chloride with Pd/H_2 and $BaSO_4/CaCO_3$ produces:
(A) Benzyl chloride (B) Benzoic acid
(C) Benzaldehyde (D) All of these

Space for rough work

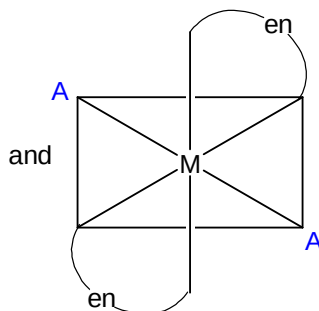
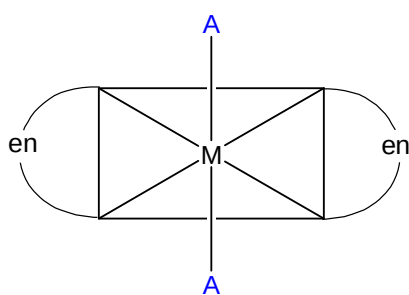
47. In the given reaction:



[A] and [B] respectively be:



48. The complexes given below are:



(A) Position Isomers
 (C) Geometrical Isomers

(B) Identical
 (D) Optical Isomers

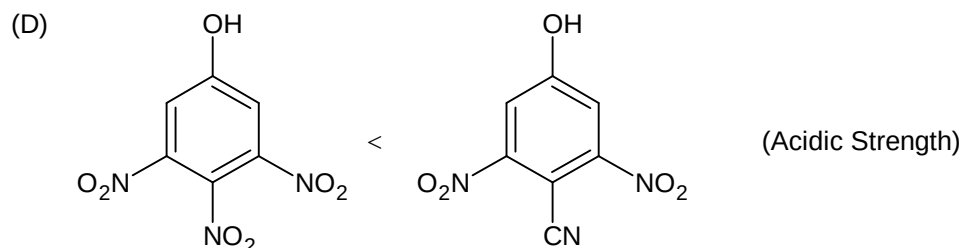
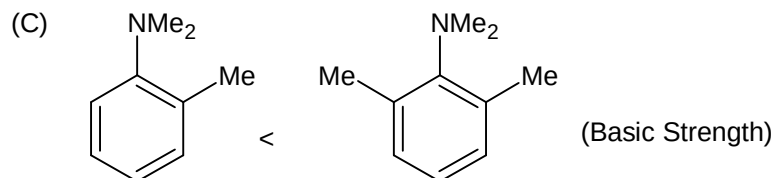
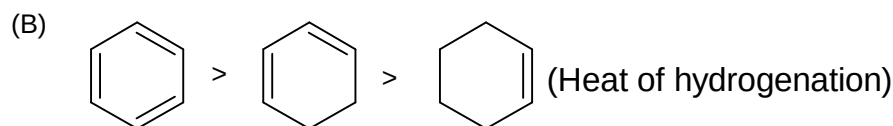
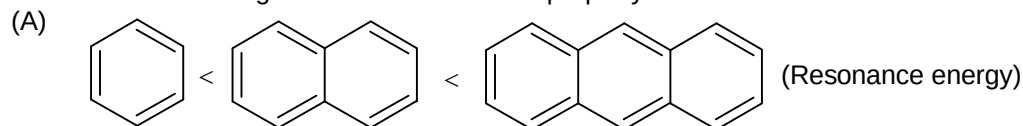
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ANY NUMBER OF OPTIONS 4, 3, 2 or 1 MAY BE CORRECT
MARKS WILL BE AWARDED ONLY IF ALL THE CORRECT OPTIONS ARE BUBBLED

49. Which of the following compound(s) will not dissolve in an alkali solution after it has undergone reaction with Hinsberg's reagent?
(A) Diethyl amine (B) Isopropyl amine
(C) N-methyl ethanamine (D) N, N-dimethyl aniline
50. $\text{CaCO}_3(\text{s}) \rightleftharpoons \text{CaO}(\text{s}) + \text{CO}_2(\text{g})$
For the above equilibrium, which is wrong?
(A) Equilibrium will shift in forward direction by addition of He at constant volume.
(B) Equilibrium will shift in forward direction by addition of He at constant pressure.
(C) Equilibrium will shift in backward direction by addition of He at constant volume.
(D) Equilibrium will shift in backward direction by addition of He at constant pressure.
51. Which of the following statements are true regarding tetrahedral voids in fcc lattice?
(A) Tetrahedral voids formed by one corner sphere and three face centre spheres in contact.
(B) Two tetrahedral voids can exist on one diagonal body line in fcc cubic lattice.
(C) Tetrahedral voids are equal to number of spheres involved in body.
(D) Number of tetrahedral voids are half of the octahedral voids.
52. Ionic conductance at infinite dilution of Al^{3+} and SO_4^{2-} ions are 60 and $80 \Omega^{-1}\text{cm}^2 \text{eq}^{-1}$, respectively. The correct details(s) regarding $\text{Al}_2(\text{SO}_4)_3$ is/are
(A) the molar conductance is $140 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$.
(B) the equivalent conductance is $140 \Omega^{-1} \text{cm}^2 \text{eq}^{-1}$.
(C) the molar conductance is $840 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$.
(D) the molar conductance is $23.33 \Omega^{-1} \text{cm}^2 \text{mol}^{-1}$.
53. The dissociation of phosgene, which occurs according to the reaction
 $\text{COCl}_2(\text{g}) \rightleftharpoons \text{CO}(\text{g}) + \text{Cl}_2(\text{g})$
is an endothermic process. Which of the following will increase the degree of dissociation of COCl_2 ?
(A) Adding Cl_2 to the system
(B) Adding helium to the system at constant pressure
(C) Decreasing the temperature of the system
(D) Reducing the total pressure

Space for rough work

54. Which of the following is correct order for the property indicated?



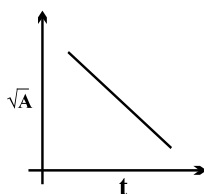
55. Assuming 2s-2p mixing is NOT operative, the diamagnetic species among the following is/are

- (A) He_2 (B) B_2
 (C) C_2 (D) N_2

56. For the reaction $\text{A} \rightarrow \text{B}$, the rate law expression is $-\frac{d[\text{A}]}{dt} = k[\text{A}]^{1/2}$. If initial concentration of $[\text{A}]$ is $[\text{A}]_0$, then

(A) The integrated rate expression is $k = \frac{2}{t}(\text{A}_0 - \text{A})$

(B) The graph of $\sqrt{\text{A}}$ Vs t will be



(C) The half life period = $\frac{K}{2[\text{A}]_0^{1/2}}$

(D) The time taken for 75% completion of reaction = $\frac{\sqrt{[\text{A}]_0}}{k}$

Space for rough work

57. Select the **correct** option(s) for an ideal gas
 (A) Most probable speed increases with increase in temperature
 (B) Fraction of particles moving with most probable speed increases with increase in temperature
 (C) Most probable speed is more for Cl_2 than H_2 at same temperature
 (D) None of these
58. Which of the following have the same number of unpaired electrons in 'd' orbitals
 (A) Cr (B) Mn
 (C) Fe^{3+} (D) Co^{3+}
59. In which reaction the product will react with 2, 4 dinitro phenylhydrazene.
- (A) $\text{H}_3\text{C}-\underset{\text{CH}_3}{\overset{\text{NH}_2}{\text{C}}}-\underset{\text{CH}_3}{\overset{\text{OH}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{NaNO}_2/\text{HCl}} \text{Product}$
- (B) $\text{H}_3\text{C}-\text{CH}_2-\underset{\text{S} \quad \text{S}}{\text{C}}-\text{CH}_3 \xrightarrow{\text{Hg}^{2+}/\text{H}_2\text{O}} \text{Product}$
- (C) $\text{Ph}-\text{C}_2\text{H}_4-\text{Ph} \xrightarrow[2. \text{HIO}_4]{1. \text{NaOH}} \text{Product}$
- (D) $\text{H}_3\text{C}-\underset{\text{H}}{\overset{\text{OH}}{\text{C}}}-\underset{\text{H}}{\overset{\text{OTS}}{\text{C}}}-\text{CH}_3 \xrightarrow{\text{OH}^-} \text{Product}$
60. Choose the correct statement/s
 (A) In sp^3d^2 hybridization participation of d_{yz} and d_{xz} orbitals is essential.
 (B) All octahedral complexes of Ni^{2+} are outer orbital complexes
 (C) All octahedral complexes of Cr^{3+} are inner orbital complexes
 (D) Temperature independent spin only magnetic moment of $[\text{MnBr}_4]^{2-}$ is 5.92 BM.

Space for rough work

ANSWERS

- | | | | |
|----------|-------------|----------------|-------------|
| 1. B | 2. C | 3. D | 4. C |
| 5. B | 6. B | 7. B | 8. A |
| 9. C | 10. D | 11. C | 12. D |
| 13. B | 14. A | 15. A | 16. D |
| 17. B | 18. C | 19. D | 20. A |
| 21. C | 22. C | 23. C | 24. A |
| 25. B | 26. B | 27. D | 28. B |
| 29. A | 30. C | 31. B | 32. C |
| 33. B | 34. B | 35. D | 36. C |
| 37. C | 38. B | 39. A | 40. D |
| 41. A | 42. A | 43. C | 44. C |
| 45. D | 46. C | 47. A | 48. B |
| 49. A, C | 50. A, C, D | 51. A, B | 52. B, C |
| 53. B, D | 54. A, C, D | 55. A, B, D | 56. B, D |
| 57. A, C | 58. A, B, C | 59. A, B, C, D | 60. B, C, D |